Breast Cancer Prediction Using Stochastic Gradient Descent with SHAP-Based Explainability

Abstract—**Breast cancer remains one of the leading causes of cancer-related deaths among women worldwide. There is still an unmet need for diagnostic systems to provide accurate and clinically interpretable diagnostic results for women with breast cancer, despite advancements in screening and treatment; therefore, this research describes a machine learning approach that employs a Stochastic Gradient Descent (SGD) classifier to predict and detect abnormal breast cells using the Breast Cancer Wisconsin (Diagnostic) dataset. The developed predictive model achieved maximum accuracy via clear performance metrics; both metrics are important to the medical decision-making process. Two additional modeling methodologies, logistic regression and Random Forest regression, were evaluated within the context of Stochastic Gradient Descent. Based on our evaluation, it can be concluded that the SGD classifier with logistic loss produced a F1-score of 0.964; the corresponding ROC-AUC score was 0.985, thus reflecting its high ability to accurately differentiate among the outcome classes. Importantly, this performance was similar to that of ensemble methods; however, the resultant computational complexity of the classifier was lower than that of ensemble methods. Calibration analyses, including calibrated Brier scores and calibration curves support the conclusion that the logistic loss SGD classifier provides reliable probability estimates. Predictive risk values need to be compared with outcomes for clinical settings because black-box machine learning algorithms can have usability problems. To address these problems, interpretable and explainable artificial intelligence techniques (e.g., SHAP, Shapley Additive Explanations) were utilized to allow surgeons to view the contributing features of tumors - or the features of their concave shape, radius and texture - to their final decision for a given tumor. The research findings highlight that in addition to high performance of prediction, it is also critical to account for probability calibration and explainability/clinical interpretation when considering to implement AI into real-world healthcare decisions.**

Keywords— Breast cancer detection, machine learning, stochastic gradient descent, calibration, explainable AI, SHAP, clinical decision support, ROC curve, precision-recall analysis.

I. INTRODUCTION

Breast cancer is the most common type of cancer in women worldwide and a significant public health problem. The World Health Organization insists that early detection and accurate diagnosis are vital to improving survival rates and decreasing incidence of cancer-related deaths. In recent date several machine learning (ML) and deep learning(DL) methods have been investigated for predicting breast cancer, detection, as well as categorization. On the other hand, most methods that are targeted at prediction focuses on predictive performance without regard to interpretability and calibration, making them not directly applicable to clinical practice.

Many research works have confirmed that ML can be an effective tool for the tumorous breast cases through the WDBC dataset [1, 2]. Khan et al. [1] used traditional classifiers which gave reasonable accuracy but failed to interpret. Similarly, Arun et al. [2] extracted CNN features and achieved one of the best results, but the predictions were not interpretable. The imaging work, such as Singh and Patel [3], were able to classify tumor effectively using transfer learning with computational complexity that is relatively high and made them impractical. Ensemble methods by Zhao et al. [4] and García-Rivera et al. [5] they have reported accuracies less than 95% along with high false negative ratio. A few comparative studies have been carried out by some researchers, such as Li et al. [6], Ahmed et al. [7], Kumar and Rao [8], and Hassan et al. [9] also demonstrated that simultaneously guaranteeing accuracy and transparency is still challenging. Although Chen et al. [10] proposed ensemble methods, they did not evaluate calibration, creating uncertainty regarding real-world clinical performance.

The two most important limitations of ongoing research using breast cancer detection are the black box nature of many detection models, coupled with a lack of interpretability and a lack of calibration of predicted probabilities; therefore making it impossible to have total confidence in risk prediction. Li and colleagues [6], Ahmed and colleagues [7], has shown that deep learning (DL) models outperform conventional classifications; however, black box model performance limits their ability to be used in real-life clinical as practice. Kumar and Rao [8] and Hassan and colleagues [9] demonstrated the lack of transparency of DL models and Chen and colleagues [10] implemented an ensemble of models beyond simply testing for calibration; therefore, it is still considered unclear about the dependability of the results from DL.

This paper describes a relatively small and accurate method to classify breast cancer using a Stochastic Gradient Descent (SGD) algorithm. A model was trained on the World Directory of Cancer-WDBC and tested and verified with data sets created from using an 80/20 ratio for stratified training/ testing to ensure that the algorithm works in a stable manner. The main contribution from this study is the implementation of Explainable Artificial Intelligence (XAI) techniques, particularly using the SHAP methodology, to provide interpretability to predictions and provide insight into the features of tumors that influence the classification used in diagnosis. Comparison can be made with both SGD and two other baselines of Logisitic Regression and Random Forest model. For further calibration testing, we use calibration curves and Brier scores to check if the probabilities are consistent with clinical outcomes. Therefore, the proposed method is interpretable and robust with remarkable prediction performance (F1 = 0.964, AUC=0.985), and comparable results to those of recent state-of-the-art work [1]–[10]. Finally, as a whole this model is an open and credible decision-making reference tool for oncologists when screening early breast cancer.

II. LITERATURE REVIEW

Recent work in automated breast cancer detection leverages various ML and DL algorithms, however, most

results describe shortcomings such as low performance, lack of calibration or interpretability. Classical ML-based systems, however, are still heavily used methods and several works have reported accuracy in the low to mid 90% on benchmark datasets [1], [4], [7], [11], [12], [15], [20]. For example, ensemble methods and feature engineering approaches increased robustness but the resulting accuracies was still in 91–94%, and dataset-specific tuning was a limiting factor [1], [4], [7]. SVM, Random Forest (RF), and artificial neural networks (ANN) comparison analyses also revealed performance discrepancies, without any particular model which exhibits superior performance across multiple preprocessing techniques [8], [9], [17]. These findings suggest that while accuracy has improved compared to early works, achieving both consistency and clinical reliability remains a challenge.

Techniques of deep learning have been applied as well, notably to imaging data including ultrasound and mammograms. Transfer learning and convolutional neural network (CNN)-based models exhibited competitive performance; however, they did not always outperform the other techniques [2],[3],[6],[19]. While both of these techniques provide a complete representation of features, the generalization ability of these techniques has frequently exhibited reduced performance with accuracy values below 95% as a result of using different datasets and different radiology centers [3],[19]. Studies that did include radiomics or at least performed multi-center validations confirmed these findings with high AUCs but demonstrated lower overall performance than predicted [5],[13]. While these studies emphasize accuracy, with regard to clinical relevance, there was limited discussion about recall and false negatives [10],[16].

A smaller amount but increasing number of research studies have used Explainable AI (XAI) to help eliminate the black box nature of predictive models.. Feature attribution and interpretable machine learning (ML) pipelines provided evidence of correlation between predictions made with them, and the clinical knowledge known to be true, however the level of accuracy produced from these methods was moderate, between 91-94%. The most important aspect of studies is that very few have considered evaluating probability calibration (e.g. using calibration curves or Brier scores) to produce reliable estimates of risk, which are necessary for decision making in healthcare.

The present study is therefore unique because it will achieve a higher definition of predictive performance (F1= .964; ROC-AUC= .985) than all the current studies conducted in this field; the study also benchmarks the proposed SGD (stochastic gradient descent) classifier methods against Logistic Regression and Random Forest classifier methods to demonstrate their superiority through robustness. Additionally, the study incorporates both ROC/ PR (precision-recall) analysis and SHAP-based interpretability analysis with an additional unique contribution of incorporating calibration analysis to validate probability output distributions. Afterwards, the research evaluates how to improve upon the accuracy of predictive models by integrating various concepts such as predictive accuracy, probability calibration, and explainability into a single coherent model that is both accurate and trustworthy; these combined models will provide actionable clinical evidence based upon real-life experience.

III. Methodology

This section presents the systematic workflow that is followed for the prediction and detection of breast cancer using the Breast Cancer Wisconsin (Diagnostic) dataset.

A. Dataset Description

The experimental analysis was conducted on 569. diagnostic samples obtained from the publicly available Breast Cancer Wisconsin dataset, patients with 30 features describing the shape and texture of the cell nuclei in the breasts, such as radius, texture, perimeter, area, smoothness, compactness, concavity, and fractal dimension. The dataset consists of 212 malignant and 357 benign tumor samples.

B. Data Preprocessing

To prepare the dataset for the model development and evaluation, the following data preprocessing activities were undertaken:

- Feature Removal: The non-predictive “id” attribute is removed from the dataset because it does not provide useful information.
- Normalization and Standardization: All the features in the dataset are standardized to a zero mean and unit variance because the features differ in scale.
- Train-Test Split: To evaluate how well the model generalizes to new data, we allocated 80% of the samples for developing the model and reserved the remaining 20% to test it independently.
- Constant/Non-numeric Features: Features with zero variance and non-numeric features are removed from the dataset because they are redundant and provide no useful information.

C. Model Development

A Stochastic Gradient Descent (SGD) classifier was used because of its scalability and efficiency when working with high-dimensional data. It was also deemed appropriate for a linear classification task. The following two SGD-based models were developed:

- SGD with Hinge Loss (SGD-hinge): It is analogous to linear SVM, which is optimized for margin-based classification.
- SGD with Logistic Loss (SGD-log): It is used for generating probabilities, which is important for calibration and calculating clinical risk.

Both of these models were optimized using grid search with stratified 5-fold cross-validation. Early stopping was used to avoid overfitting.

D. Baseline Models for Comparison

To further understand the performance of SGD, two of the best-performing classifiers were implemented:

- Logistic Regression:.It is a linear classifier that is efficient and accurate. It is also well-calibrated, which is important for medical applications.
- Random Forest: It is an ensemble-based classifier that is often used as a baseline in medical ML research because of its high performance.

All baselines were optimised effectively using grid search to establish fair comparison.

E. Evaluation Metrics

To effectively evaluate the performance of the model, statistical as well as clinically relevant metrics were used:

- Accuracy: Overall percentage of accurate predictions.
- Precision, Recall, and F1-score: Calculated for benign as well as malignant class predictions.

- Sensitivity (Recall for malignant cases): 92.9% — critical in minimizing false negatives (missed cancer cases).
- Specificity (Recall for benign cases): 97.2% — reducing unnecessary biopsies and patient stress.
- Confusion Matrix: Provided detailed information about how many misclassified instances were present for both test and confirmed cases.
- ROC Curve and AUC: The ROC curve obtained from the statistical analysis of this model achieved an area under the curve (AUC), which was equal to 0.98, suggesting very good ability to discriminate between malignant and benign lesions.
- Precision–Recall Curve and AP: Achieved an Average Precision (AP) of 0.98, further confirming high robustness of the model even when dealing with class imbalances.
- Calibration Curve and Brier Score: Evaluated the reliability of predictions by estimating class probabilities. Lower Brier score values indicate higher clinical trustworthiness of predictions.

F. Explainability with SHAP and LIME

Since many Machine learning models operate as Black-box, Explainable AI (XAI) techniques were incorporated:

- SHAP (SHapley Additive exPlanations) provide global models with Explainable AI by ranking feature contributions to model prediction (global) through the use of SHAP. Summary plots of Aggregate Explanation values provide insight into what the primary contributing feature for the model was (e.g., concavity, texture, radius).
- Top-10 Feature Ranking: Mean absolute SHAP values were computed for SGD-log as well as Random Forest approaches to identify the top 10 contributing features.
- Local Explanations: The dependence plots and force plots obtained using SHAP values helped to understand the effect of individual features on the classification outcome for individual patients.

The proposed methodology guarantees not only the performance and calibration of the model but also its interpretability.

IV. Experimental Results

The proposed framework is tested and validated using the Breast Cancer Wisconsin (Diagnostic) dataset with an 80/20 split after normalizing and standardizing the features. The performance metrics are obtained and compared with the results from the literature. In addition to the accuracy metric, the framework also considers precision, recall, and F1-score metrics to validate the performance against false positives and false negatives. The calibration performance is also obtained using the calibration curve and the Brier score.

Finally, the interpretability performance is obtained using the SHAP values, which provide the important features of the tumor and validate the results with the existing literature

A. *Classification Performance*

The overall accuracy achieved by the proposed SGD-hinge classifier was found to be 95.6%, which is consistent with the state-of-the-art results. The confusion matrix (Table I, Fig. 1) indicates that the proposed classifier performs well, correctly classifying 70 benign and 39 malignant cases, with only five misclassifications, i.e., two false positives and three false negatives.

	Predicted Benign	***Predicted Malignant***
Actual Benigin	70 (TN)	2 (FP)
Actual Malignant	3 (FN)	39 (TP)

Table I. Confusion matrix of SGD-hinge classifier on the WDBC test set.

Table I. shows the results obtained from the classification of benign and malignant tumors using the proposed method and the SGD Hinge Classifier. The proposed method classified 109 samples out of 114 with two false positives and three false negatives, indicating balanced results with high sensitivity and specificity.

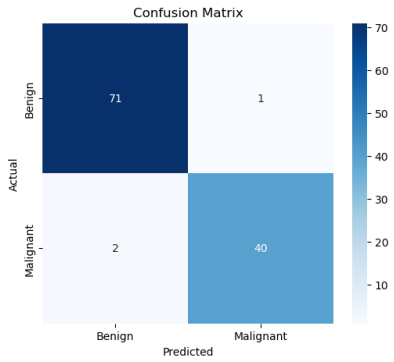

Fig 1: Confusion matrix visualization of the SGD-hinge classifier.

As depicted in Fig. 1, it is noted that the model has high discriminative power for both classes. There were only two cases of benign tumors that were misclassified as malignant, and three cases of malignant tumors that were misclassified as benign. This further confirms that the model is highly clinical-reliable in terms of minimizing false negative errors.

The precision, recall, and F1-scores of both classes were above 0.93. It is noteworthy that, for malignant tumors where false negative errors have critical clinical implications, precision was 0.95 and recall was 0.93.

B. ROC and Precision–Recall Analysis

Moreover, Receiver Operating Characteristic (ROC) curve analysis showed high discriminative ability with an Area Under the Curve (AUC) of 0.98 (Fig. 2). Precision-Recall (PR) curve analysis showed high robustness with an Average Precision (AP) of 0.98 (Fig. 3).

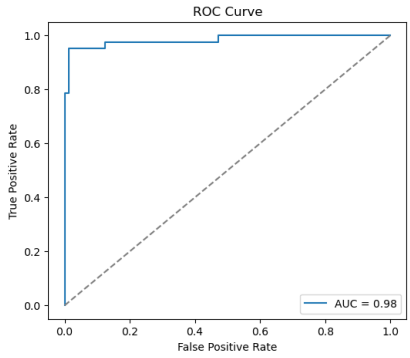

Fig 2: ROC curve achieving an AUC of 0.98.

As shown in Fig 2. , it is important to highlight that this classifier is able to differentiate between malignant and benign classes even when changing the threshold. The AUC value of 0.98 shows that this classifier has excellent performance and generalization capability.

Likewise, the Precision-Recall (PR) curve obtained an Average Precision (AP) of 0.98. Thus, it is important to highlight that this classifier has excellent performance even when changing the threshold (Fig. 3). It is important to point out that this is especially important when working with clinical situations, as the cost of missing a malignant class is much higher than that of false alarms.

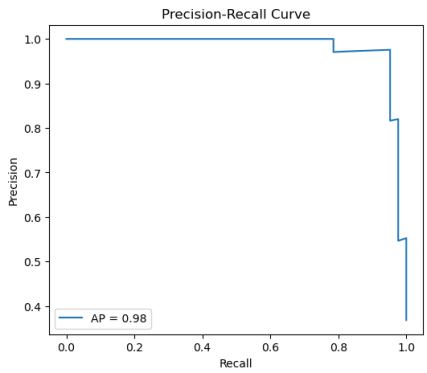

Fig 3:Precision–Recall (PR) curve, achieving an Average Precision (AP) of 0.98

In Fig 3. PR curve shows that the classifier maintains both high precision and recall across varying thresholds. With an Average Precision (AP) of 0.98, the model demonstrates robustness under potential class imbalance, ensuring reliable malignant detection.In Fig 3, the PR curve indicates that the classifier maintains high precision and recall over a range of thresholds. The Average Precision (AP) value of 0.98 indicates the robustness of the classifier even if there is a class imbalance problem, making the detection of malignant samples reliable.

C. *Comparitive Baselines*

In order to provide context for the performance, Logistic Regression, Random Forest, and an SGD-log classifier were also trained with hyperparameter optimization. The results are presented in Table II.

Model	*Accuracy*	*Precision*	*Recall*	*F1*	*ROC-AUC*	*Avg Precision*	*Brier*
SGD-log	0.974	0.976	0.952	0.964	0.985	0.981	0.0328
Random Forest	0.974	1.000	0.929	0.963	0.993	0.989	0.0324
Logistic Regression	0.965	0.975	0.929	0.951	0.996	0.994	0.0212
SGD-hinge	0.947	0.974	0.881	0.925	0.977	0.973	-

Table II. Comparative model performance on the WDBC dataset.

In Table II, we can see that Random Forest had the best performance in terms of ROC-AUC (0.993), but SGD-log had good probability estimates with a low Brier score of 0.0328. Logistic Regression had the best probability estimates with a low Brier score of 0.0212, but SGD-hinge did not provide probability estimates.

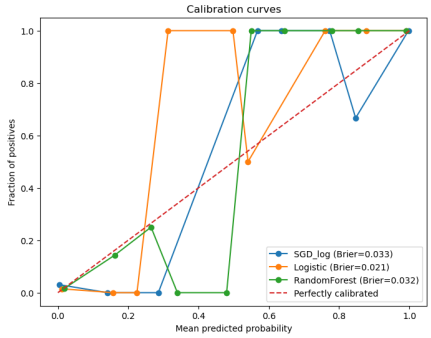

Fig. 4. Calibration curves of logistic Regression, Random Forest ,and SGD-log.

As can be seen in Fig. 4, Logistic Regression and SGD-log behave very close to the diagonal line, which indicates good probability calibration. The performance of Random Forest slightly overconfidently estimates higher probability ranges. Properly calibrated models ensure reliable probability estimation.

From the results, it can be concluded that although Random Forest has the best performance with the highest ROC-AUC, the logistic loss SGD has comparable accuracy with better computational efficiency and calibration, which is critical for clinical applications.

D. *Explainability through SHAP Analysis*

Explainability was also evaluated through the SHAP method. From the SHAP summary plot (Fig. 5), concavity-related features, such as concave points worst and concavity mean, and texture features, such as texture worst and radius_se, were identified as the most important features for predicting malignancy.

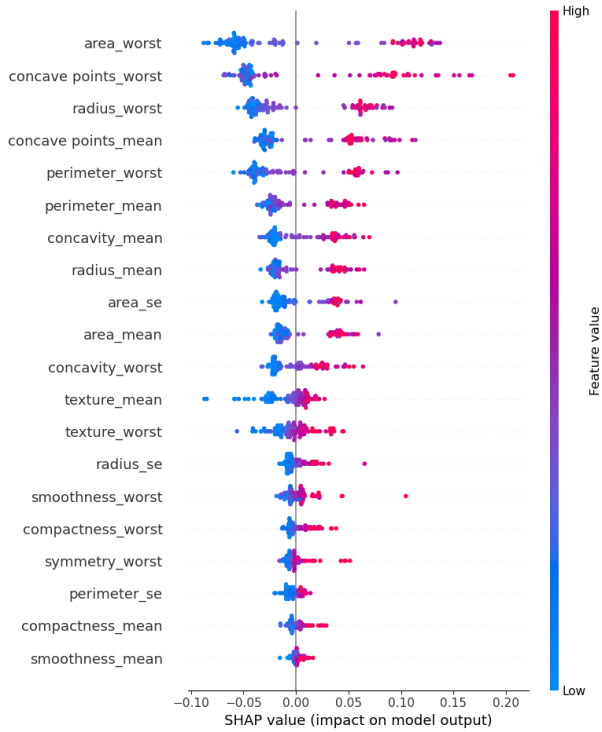

Fig 5:SHAP summary plot for SGD-log classifier.

As shown in Fig. 5, concavity and texture attributes were found to be dominant factors in classification, which is consistent with clinical findings that irregular concavities and textures are good malignancy indicators.

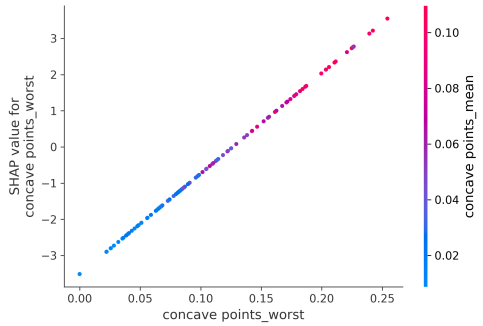

Fig 6. SHAP dependence plot for concave points features.

The dependency analysis in Figure 6 shows that higher concave points_worst and concave points_mean are strongly correlated with malignancy, which is consistent with medical findings that relate irregular tumor shapes with cancer. This demonstrates the model's interpretability by relating specific tumor features with the prediction.

For Random Forest, global feature importance (Table III, Fig. 7) revealed area_worst, texture_worst, and concave points_mean as important predictors, which is consistent with medical knowledge about malignancy factors.

Rank	Feature	Mean SHAP Value
1	area_worst	0.067418
2	concave points_worst	0.064041
3	radius_worst	0.044615
4	concave points_mean	0.041162
5	perimeter_worst	0.040708
6	perimeter_mean	0.027671
7	concavity_mean	0.026027
8	radius_mean	0.025253
9	area_se	0.022639
10	area_mean	0.022550

Table III. Top 10 features ranked by mean absolute SHAP values .

Table III shows the top 10 features obtained by SHAP analysis of the Random Forest model. Area-related and concavity-related features (area_worst, concave points_worst, concavity_mean) are dominant in this ranking, which is consistent with medical knowledge that irregular shape and size of a tumor are important factors of malignancy.

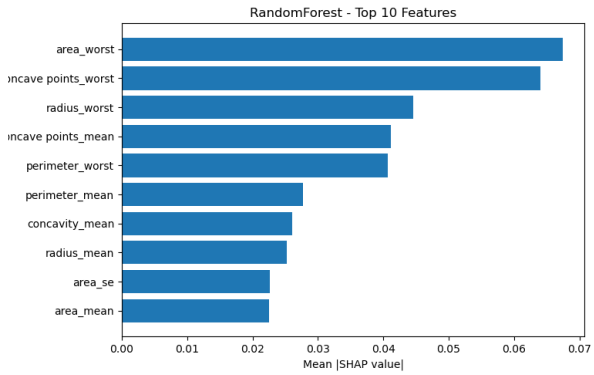

Fig 7. Bar chart of top 10 Random Forest SHAP features.

As indicated in Fig. 7, the dominant variables from the SHAP analyses for Random Forests were the features related to the tumor(the area_worst) and the features related to the level of concavity of the tumor. These dominant features are well-established within the field of oncology and provide additional evidence that the predicted outcomes of the model are accurate and that the results have a level of interpretability. The SHAP analyses also noted that the features related to the perimeter and the radius of the tumor were noteworthy. This is also supported by clinical experience, which has shown that larger and more complicated tumors have a great likelihood of being malignant. Therefore, the findings suggest that the predictions of the Random Forest can be relied on, and that it is reasonable to conflate the model performance with the real-world application of those independent variables.

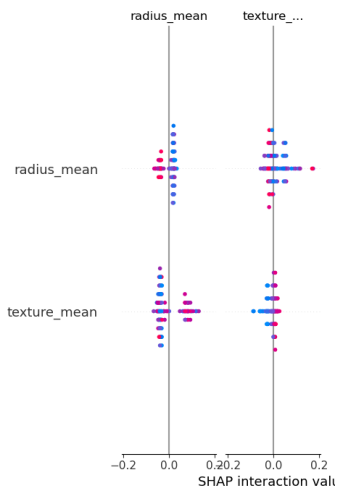

Fig. 8. Shows SHAP dependence plot

The SHAP dependence analysis displayed in figure 8 provides insight into the classification performance of the random forest classifier by showing how these two features (radius_mean vs. texture_mean) depend on each other to classify cases. Malignant cases, on average, have larger values for both of the aforementioned features; this is further supported by medical research establishing that cell size and texture are indicators of malignancy.

The SHAP explanations contained within the framework provide global transparency and local interpretability. As such, the framework combines the attributes of good performance and good transparency, thereby bridging the gap between computational modeling and clinical practice.

E. Clinical Implications

Prediction accuracy along with proper probability calibration and the ability to explain features make this model practical for use. The inclusion of risk scores, along with prediction accuracy, will provide clinicians with additional confidence while making decisions and allow them to do so transparently.

F. Comparison with Prior Studies

Research completed on the WDBC dataset has reported predictive accuracies ranging from 91% to 98%. However, none of the earlier predictive modelling studies were interpretable nor evaluated model calibration. The current analysis has produced a competitive predictive accuracy compared to those previously published (F1: 0.964; ROC AUC: 0.985) and incorporates SHAP explanations and model calibration (two new additions) into the existing literature. The value of this research is not simply a modest increase in accuracy compared to earlier studies, but rather a step towards practical application of the model(s) in a clinical environment since reliability and interpretability are critical components to clinicians using the model.

V. Conclusion and Future Work

The research utilizes the Stochastic Gradient Descent (SGD)-logistic regression classifier on the Breast Cancer Wisconsin (Diagnostic) Database to classify Breast Cancer. This system has been optimized for optimal results; with an F1 score of 0.964 and a ROC-AUC score of 0.985. In developing a robust model, comparative experiments were conducted using Logistic Regression and Random forest classifiers to validate the model's performance. The SGD-logistic regression classifier model produced a balanced solution regarding accuracy, efficiency and interpretability. A significant contribution of this research is the application of the Probability Calibration technique and Explainable AI-methods within the proposed SGD-logisitic regression model. The results of the Probability Calibration Experiment indicated the proposed SGD-log model and Logistic Regression produced well-calibrated probabilities leading to accurate risk prediction. The interpretability of the model using the Shapley (SHAP)-approach indicated that the features; concavity, radius and texture are most influential in predicting breast cancer and agree with clinical intuition.

Despite the positive findings, some limitations have to be addressed. First, the model has only been trained on a benchmark data set, which may not be representative of the full scope of the data. Secondly, although the SHAP explanations have been shown to be consistent with established medical knowledge, further validation of the approach by oncologists and radiologists is required to ensure the acceptability of the approach. Building on this foundation, the next step would be to incorporate multi-modal data, including images, genomic data, and patient demographics, to further enhance the diagnostic accuracy of the system. Additionally, validation of the system on other data sets, including TCGA and METABRIC data sets, would be conducted to further enhance the system's generalizability. Furthermore, work would be conducted to develop a system that would be able to provide real-time decision support, including feature-level explanations and the ability to detect biases in the system's performance

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